# SUMMARY OF SAFETY AND EFFECTIVENESS DATA (SSED)

## I. GENERAL INFORMATION

Device Generic Name: Drug-Eluting Coronary Stent System

Device Trade Name: Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System

Device Procode: NIQ

Applicant's Name and Address: Medtronic Vascular
3576 Unocal Place
Santa Rosa, CA 95403

Date(s) of Panel Recommendation: None

Premarket Approval Application (PMA) Number: P160043/S001

Date of FDA Notice of Approval: November 16, 2017

The original PMA (P160043) was approved on April 28, 2017 and is indicated for improving coronary luminal diameters in patients, including those with diabetes mellitus, with symptomatic ischemic heart disease due to *de novo* lesions of length ≤ 35 mm in native coronary arteries with reference vessel diameters of 2.25 mm to 5.0 mm. The SSED to support the indication is available on the CDRH website (https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P160043) and is incorporated by reference here. The current supplement was submitted to expand the indication for the Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System to include reference vessel diameters of 2.0 mm.

## II. INDICATIONS FOR USE

The Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System is indicated for improving coronary luminal diameters in patients, including those with diabetes mellitus, with symptomatic ischemic heart disease due to *de novo* lesions of length ≤ 35 mm in native coronary arteries with reference vessel diameters of 2.0 mm to 5.0 mm.

## III. CONTRAINDICATIONS

The Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System is contraindicated for use in:

- Patients with known hypersensitivity or allergies to aspirin, heparin, bivalirudin, clopidogrel, prasugrel, ticagrelor, ticlopidine, drugs such as zotarolimus, tacrolimus, sirolimus, everolimus, or similar drugs or any other analogue or derivative.

- Patients with a known hypersensitivity to the cobalt-based alloy (cobalt, nickel, chromium, and molybdenum) or platinum-iridium alloy.
- Patients with a known hypersensitivity to the BioLinx polymer or its individual components

Coronary artery stenting is contraindicated for use in:

- Patients in whom anti-platelet and/or anticoagulation therapy is contraindicated.
- Patients who are judged to have a lesion that prevents complete inflation of an angioplasty balloon or proper placement of the stent or stent delivery system.

## IV. WARNINGS AND PRECAUTIONS

The warnings and precautions can be found in the Resolute Onyx Zotarolimus-Eluting Coronary Stent System labeling.

## V. DEVICE DESCRIPTION

The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) is a device/drug combination product comprised of the following device components:

- A Resolute Onyx™ Coronary Stent and delivery system. The delivery system is available in a rapid exchange (RX) and an over-the-wire (OTW) configuration.
- A drug/polymer coating component, which consists of a formulation of zotarolimus contained in a BioLinx polymer.

The characteristics of the Resolute Onyx™ product are described in **Table 1**.

**Table 1: Device Component Description and Nominal Dimensions**

| Component | Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System Rapid Exchange and Over-the-Wire Delivery Systems | | | |
|---|---|---|---|---|
| | **Stent Design 1 (Small Vessel)** | **Stent Design 2 (Medium Vessel)** | **Stent Design 3 (Large Vessel)** | **Stent Design 4 (Extra Large Vessel )** |
| Available Stent Diameters (mm) | 2.0, 2.25, 2.5 | 2.75, 3.0 | 3.5, 4.0 | (RX Only) – 4.5, 5.0 |
| Available Stent Lengths (mm) | 8, 12, 15, 18, 22, 26, 30, 34*, 38*<br>* 34, 38 mm lengths not available in 2.0 mm | 8, 12, 15, 18, 22, 26, 30, 34, 38 | 8, 12, 15, 18, 22, 26, 30, 34, 38 | (RX Only) – 12, 15, 18, 22, 26, 30 |

**Table 1: Device Component Description and Nominal Dimensions**

| Component | | Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System Rapid Exchange and Over-the-Wire Delivery Systems | | | |
|---|---|---|---|---|---|
| | | **Stent Design 1 (Small Vessel)** | **Stent Design 2 (Medium Vessel)** | **Stent Design 3 (Large Vessel)** | **Stent Design 4 (Extra Large Vessel )** |
| Stent Material and Geometry | | A continuous sinusoid pattern stent manufactured from a composite metal material, consisting of a cobalt-based alloy shell conforming to ASTM F562 and a platinum-iridium alloy core conforming to ASTM B684. | | | |
| Drug Component | | A coating of polymers loaded with zotarolimus in a formulation applied to the entire surface of the stent at a dose of approximately 1.6 µg/mm$^2$ which results in a maximum nominal drug content of 317 µg on the stent with the largest surface area (4.0 x 38 mm). | | | |
| Delivery System Working Length | | 140 cm | | | |
| Delivery System Luer Adapter Ports | RX | Single access port to the inflation lumen. A guidewire exit port is located approximately 25 cm from the tip. Designed for guidewire less than or equal to 0.014 inch (0.36 mm). | | | |
| | OTW | Y-Connector with side arm for access to balloon inflation/deflation lumen. Straight arm is continuous with shaft inner lumen designed for guidewire less than or equal to 0.014 inch (0.36 mm). | | | |
| Stent Delivery Balloon | | Single-layer Pebax balloon, wrapped over an inner member tubing with 2 radiopaque marker bands to locate the stent edges. | | | |
| Balloon Inflation Pressure | | Nominal Inflation Pressure: 12 ATM (1216 kPa)<br>Rated Burst Pressure: 2.0-4.0mm = 18 ATM (1824 kPa), RX only: 4.5-5.0mm = 16 ATM (1621kPa) | | | |
| Minimum Guide Catheter Inner Diameter | | 5 F (1.42 mm, 0.056 in) | | | |
| Catheter Shaft Outer Diameter | RX | Proximal Shaft OD, 2.0-5.0mm: 2.1 F (0.69 mm)<br>Distal Shaft OD, 2.0-4.0mm: 2.7 F (0.91 mm)<br>Distal Shaft OD, 4.5 and 5.0mm: 3.2 F (1.07 mm) | | | |
| | OTW | Proximal Shaft OD: 3.4 F (1.12 mm)<br>Distal Shaft OD: 2.7 F (0.91 mm) | | | |

## A. Device Component Description

The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) consists of a balloon-expandable, intracoronary, drug-eluting stent (DES) premounted on a stent delivery system (RX or OTW). The Resolute Onyx™ coronary stent is manufactured from a composite material of cobalt alloy and platinum-iridium alloy and is formed from a single wire bent into a continuous sinusoid pattern and then laser fused back onto itself. The Resolute Onyx™ cornary stent is coated with a Parylene C primer, a Biolinx polymer, and the active pharmaceutical ingredient (API), zotarolimus with a nominal drug dose density of approximately 1.6 µg/mm$^2$. The stents

are available in multiple lengths and diameters. The delivery system has two radiopaque markers to aid in the placement of the stent during fluoroscopy and is compatible with 0.014-inch (0.36-mm) guidewires and 1.42-mm (5-Fr/0.056-in) minimum inner diameter guide catheters. The stent is crimped on various sizes of delivery catheter balloons, which range from 2.0 mm to 5.0 mm. See Table 1, above, for full list of diameter ranges available on each delivery system (RX and OTW).

**B. Drug Component Description**

The drug coating of the Resolute Onyx™ System consists of the drug zotarolimus (the active ingredient) and BioLinx® polymer system (the inactive ingredient).

**B1. Active Ingredient: Zotarolimus**

The active pharmaceutical ingredient utilized in Resolute Onyx™ is zotarolimus. It is a tetrazole-containing macrocyclic immunosuppressant.

The Chemical name of zotarolimus is: [3S-[3R*[S*(1R*,3S*,4R*)],6S*, 7E,9S*,10S*,12S*,14R*,15E,17E,19E, 21R* ,23R*, 26S*,27S*,34aR*]]-9,10,12,13,14,21,22,23,24,25,26,27, 32,33,34,34a-hexadecahydro-9,27-dihydroxy-3-[2-[3-methoxy-4-(1H-tetrazoyl-1-yl)cyclohexyl]-1-methylethyl]-10,21-dimethoxy-6,8,12,14,20,26-hexamethyl-23,27-epoxy-3H-pyrido[2,1-c] [1,4]oxaazacyclohentriacontine-1,5,11,28,29(4H,6H,31H)-pentone.

The chemical structure of zotarolimus is shown in **Figure 1**.

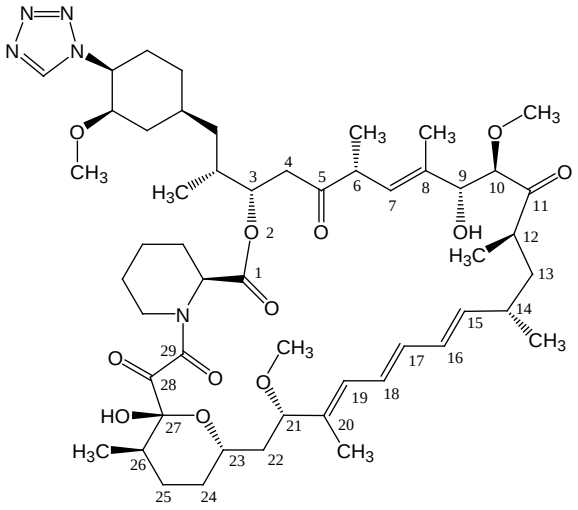


**Figure 1: Chemical Structure of Zotarolimus**

Zotarolimus has extremely low water solubility and is a lipophilic compound that is freely soluble in Propylene glycol, Acetone, Toluene, Acetonitrile, Ethanol, Benzyl alcohol and DMSO. The molecular formula of zotarolimus is $C_{52}H_{79}N_5O_{12}$ and its molecular weight is 966.2.

Zotarolimus does not have any ionizable group(s) in the physiological pH range; therefore, its solubility is expected to be unaltered in this range.

## B2. Inactive Ingredient

Biolinx Polymer

Resolute Onyx™ coronary stent is covered with a coating that consists of a blend of the drug zotarolimus and the Biolinx polymer system. BioLinx is a blend of the Medtronic proprietary components C10 and C19, and PVP (polyvinyl pyrrolidone).

The structural formula of the BioLinx polymer subunits is show in **Figure 2**.

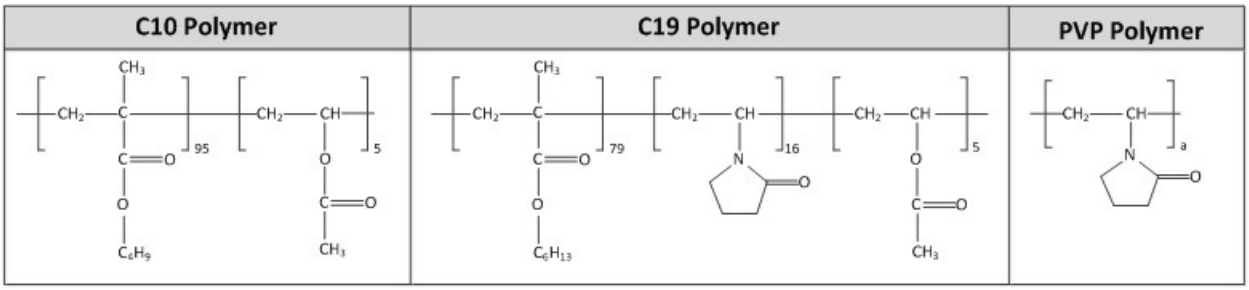



**Figure 2: Chemical Structure of Biolinx Polymer Sub-units**

**Table 2: Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System Product Matrix and Nominal Zotarolimus Content**

| Product Number Resolute Onyx RX | Product Number Resolute Onyx OTW | Nominal Expanded Stent ID RX (mm) | Nominal Unexpanded Stent Length RX & OTW (mm) | Nominal Zotarolimus Content RX (µg) | Nominal Zotarolimus Content OTW (µg) |
|---|---|---|---|---|---|
| RONYX20008UX | RONYX20008W | 2.0 | 8 | 51 | 51 |
| RONYX22508UX | RONYX22508W | 2.25 | | 51 | 51 |
| RONYX25008UX | RONYX25008W | 2.5 | | 51 | 51 |
| RONYX27508UX | RONYX27508W | 2.75 | | 67 | 67 |
| RONYX30008UX | RONYX30008W | 3.0 | | 67 | 67 |
| RONYX35008UX | RONYX35008W | 3.5 | | 77 | 77 |
| RONYX40008UX | RONYX40008W | 4.0 | | 77 | 77 |
| RONYX20012UX | RONYX20012W | 2.0 | 12 | 70 | 70 |
| RONYX22512UX | RONYX22512W | 2.25 | | 70 | 70 |
| RONYX25012UX | RONYX25012W | 2.5 | | 70 | 70 |
| RONYX27512UX | RONYX27512W | 2.75 | | 94 | 94 |
| RONYX30012UX | RONYX30012W | 3.0 | | 94 | 94 |
| RONYX35012UX | RONYX35012W | 3.5 | | 108 | 108 |
| RONYX40012UX | RONYX40012W | 4.0 | | 108 | 108 |
| RONYX45012UX | Not Available | 4.5 | | 132 | Not Available |
| RONYX50012UX | Not Available | 5.0 | | 132 | Not Available |
| RONYX20015UX | RONYX20015W | 2.0 | 15 | 85 | 85 |
| RONYX22515UX | RONYX22515W | 2.25 | | 85 | 85 |

**Table 2: Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System Product Matrix and Nominal Zotarolimus Content**

| Product Number Resolute Onyx RX | Product Number Resolute Onyx OTW | Nominal Expanded Stent ID RX (mm) | Nominal Unexpanded Stent Length RX & OTW (mm) | Nominal Zotarolimus Content RX (µg) | Nominal Zotarolimus Content OTW (µg) |
|---|---|---|---|---|---|
| RONYX25015UX | RONYX25015W | 2.5 | | 85 | 85 |
| RONYX27515UX | RONYX27515W | 2.75 | | 117 | 117 |
| RONYX30015UX | RONYX30015W | 3.0 | | 117 | 117 |
| RONYX35015UX | RONYX35015W | 3.5 | | 132 | 132 |
| RONYX40015UX | RONYX40015W | 4.0 | | 132 | 132 |
| RONYX45015UX | Not Available | 4.5 | | 158 | Not Available |
| RONYX50015UX | Not Available | 5.0 | | 158 | Not Available |
| RONYX20018UX | RONYX20018W | 2.0 | | 104 | 104 |
| RONYX22518UX | RONYX22518W | 2.25 | | 104 | 104 |
| RONYX25018UX | RONYX25018W | 2.5 | | 104 | 104 |
| RONYX27518UX | RONYX27518W | 2.75 | | 140 | 140 |
| RONYX30018UX | RONYX30018W | 3.0 | 18 | 140 | 140 |
| RONYX35018UX | RONYX35018W | 3.5 | | 156 | 156 |
| RONYX40018UX | RONYX40018W | 4.0 | | 156 | 156 |
| RONYX45018UX | Not Available | 4.5 | | 188 | Not Available |
| RONYX50018UX | Not Available | 5.0 | | 188 | Not Available |
| RONYX20022UX | RONYX20022W | 2.0 | | 127 | 127 |
| RONYX22522UX | RONYX22522W | 2.25 | | 127 | 127 |
| RONYX25022UX | RONYX25022W | 2.5 | | 127 | 127 |
| RONYX27522UX | RONYX27522W | 2.75 | | 171 | 171 |
| RONYX30022UX | RONYX30022W | 3.0 | 22 | 171 | 171 |
| RONYX35022UX | RONYX35022W | 3.5 | | 186 | 186 |
| RONYX40022UX | RONYX40022W | 4.0 | | 186 | 186 |
| RONYX45022UX | Not Available | 4.5 | | 227 | Not Available |
| RONYX50022UX | Not Available | 5.0 | | 227 | Not Available |
| RONYX20026UX | RONYX20026W | 2.0 | | 146 | 146 |
| RONYX22526UX | RONYX22526W | 2.25 | | 146 | 146 |
| RONYX25026UX | RONYX25026W | 2.5 | | 146 | 146 |
| RONYX27526UX | RONYX27526W | 2.75 | | 198 | 198 |
| RONYX30026UX | RONYX30026W | 3.0 | 26 | 198 | 198 |
| RONYX35026UX | RONYX35026W | 3.5 | | 221 | 221 |
| RONYX40026UX | RONYX40026W | 4.0 | | 221 | 221 |
| RONYX45026UX | Not Available | 4.5 | | 265 | Not Available |
| RONYX50026UX | Not Available | 5.0 | | 265 | Not Available |
| RONYX20030UX | RONYX20030W | 2.0 | | 168 | 168 |
| RONYX22530UX | RONYX22530W | 2.25 | | 168 | 168 |
| RONYX25030UX | RONYX25030W | 2.5 | | 168 | 168 |
| RONYX27530UX | RONYX27530W | 2.75 | | 225 | 225 |
| RONYX30030UX | RONYX30030W | 3.0 | 30 | 225 | 225 |
| RONYX35030UX | RONYX35030W | 3.5 | | 252 | 252 |
| RONYX40030UX | RONYX40030W | 4.0 | | 252 | 252 |
| RONYX45030UX | Not Available | 4.5 | | 304 | Not Available |

**Table 2: Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System Product Matrix and Nominal Zotarolimus Content**

| Product Number Resolute Onyx RX | Product Number Resolute Onyx OTW | Nominal Expanded Stent ID RX (mm) | Nominal Unexpanded Stent Length RX & OTW (mm) | Nominal Zotarolimus Content RX (µg) | Nominal Zotarolimus Content OTW (µg) |
|---|---|---|---|---|---|
| RONYX50030UX | Not Available | 5.0 | | 304 | Not Available |
| RONYX22534UX | RONYX22534W | 2.25 | 34 | 187 | 187 |
| RONYX25034UX | RONYX25034W | 2.5 | | 187 | 187 |
| RONYX27534UX | RONYX27534W | 2.75 | | 257 | 257 |
| RONYX30034UX | RONYX30034W | 3.0 | | 257 | 257 |
| RONYX35034UX | RONYX35034W | 3.5 | | 282 | 282 |
| RONYX40034UX | RONYX40034W | 4.0 | | 282 | 282 |
| RONYX22538UX | RONYX22538W | 2.25 | 38 | 206 | 206 |
| RONYX25038UX | RONYX25038W | 2.5 | | 206 | 206 |
| RONYX27538UX | RONYX27538W | 2.75 | | 284 | 284 |
| RONYX30038UX | RONYX30038W | 3.0 | | 284 | 284 |
| RONYX35038UX | RONYX35038W | 3.5 | | 317 | 317 |
| RONYX40038UX | RONYX40038W | 4.0 | | 317 | 317 |

### C. Mechanism of Action

*In vitro*, zotarolimus inhibited growth factor-induced proliferation of human coronary artery smooth muscle cells, and also demonstrated binding affinity with FKBP-12 (binding protein). The suggested mechanism of action of zotarolimus is to bind to FKBP12, leading to the formation of a trimeric complex with the protein kinase mTOR (mammalian target of rapamycin), inhibiting its activity. Inhibition of mTOR activity leads to inhibition of cell cycle progression from the G1 to the S phase. **Table 2**, above, lists the nominal drug content present on each product included in the Resolute Onyx™ matrix.

## VI. ALTERNATIVE PRACTICES AND PROCEDURES

There are several other alternatives for the correction of coronary artery disease including exercise, diet, smoking cessation counseling, drug therapy, percutaneous coronary interventions (such as balloon angioplasty, atherectomy, and replacement with bare metal stents, coated stents, and other drug eluting stents), and coronary artery bypass surgery (CABG). Each alternative has its own advantages and disadvantages. A patient should fully discuss these alternatives with his/her physician to select the method that best meets expectations and lifestyle.

## VII. MARKETING HISTORY

The Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System is commercially available in the following countries:

| | | | |
|---|---|---|---|
| Albania | El Salvador | Lebanon | Reunion |
| Algeria | Estonia | Lithuania | Romania |
| Argentina | Fiji | Luxembourg | Russian Federation |
| Armenia | Finland | Macedonia | Saudi Arabia |
| Australia | France | Malaysia | Serbia |
| Austria | Germany | Malta | Singapore |
| Azerbaijan | Greece | Martinique | Slovakia |
| Bahrain | Guatemala | Mexico | Slovenia |
| Bangladesh | Honduras | Moldova | South Africa |
| Belgium | Hong Kong | Montenegro | Spain |
| Bolivia | Hungary | Morocco | Sri Lanka |
| Bosnia and Herzegovina | Iceland | Namibia | Sweden |
| Botswana | India | Nepal | Switzerland |
| Brunei Darussalam | Iran | Netherlands | Taiwan |
| Bulgaria | Iraq | New Zealand | Tajikistan |
| Canary Islands | Ireland | Nicaragua | Tanzania |
| Columbia | Israel | Norway | Thailand |
| Costa Rica | Italy | Oman | Trinidad And Tobago |
| Croatia | Jordan | Pakistan | Tunisia |
| Cyprus | Kazakhstan | Panama | Turkey |
| Czech Republic | Kenya | Paraguay | United Arab Emirates |
| Denmark | Korea, Republic Of | Philippines | United Kingdom |
| Dominican Republic | Kuwait | Poland | United States |
| Ecuador | Kyrgyzstan | Portugal | Venezuela |
| Egypt | Latvia | Qatar | Yemen |

The device has not been withdrawn from marketing for any reason related to its safety or effectiveness.

## VIII. POTENTIAL ADVERSE EFFECTS OF THE DEVICE ON HEALTH

Below is a list of the potential adverse effects (e.g., complications) associated with the use of the device. Adverse events (in alphabetical order) which may be associated with coronary stent use in native coronary arteries include, but are not limited to:

- Abrupt vessel closure
- Access site pain, hematoma, or hemorrhage
- Allergic reaction (to contrast, antiplatelet therapy, stent material, or drug and polymer coating)
- Aneurysm, pseudoaneurysm, or arteriovenous fistula (AVF)
- Arrhythmias, including ventricular fibrillation
- Balloon rupture
- Bleeding
- Cardiac tamponade
- Coronary artery occlusion, perforation, rupture, or dissection
- Coronary artery spasm

- Death
- Embolism (air, tissue, device, or thrombus)
- Emergency surgery: peripheral vascular or coronary bypass
- Failure to deliver the stent
- Hemorrhage requiring transfusion
- Hypotension/hypertension
- Incomplete stent apposition
- Infection or fever
- Myocardial Infarction (MI)
- Pericarditis
- Peripheral ischemia/peripheral nerve injury
- Renal failure
- Restenosis of the stented artery
- Shock/pulmonary edema
- Stable or unstable angina
- Stent deformation, collapse, or fracture
- Stent migration or embolization
- Stent misplacement
- Stroke/transient ischemic attack
- Thrombosis (acute, subacute, or late)

Adverse events that have been associated with the intravenous injection of zotarolimus in humans include but are not limited to:

- Anemia
- Diarrhea
- Dry Skin
- Headache
- Hematuria
- Infection
- Injection site reaction
- Pain (abdominal, arthralgia, injection site)
- Rash

Potential adverse events related to BioLinx polymer include but are not limited to:

- Allergic reaction
- Focal inflammation at the site of stent implantation
- Restenosis of the stented artery

For the specific adverse events that occurred in the RESOLUTE ONYX 2.0 mm Clinical Study, please see **Section X** below.

## IX. SUMMARY OF NONCLINICAL STUDIES

A series of non-clinical laboratory studies related to the Resolute family of products were performed and the pertinent data is being leveraged from the previously approved PMAs P110013 and P160043. Because these previously collected data sufficiently represent the performance of the device for the new indications for use, no new non-clinical testing was conducted.

## X. SUMMARY OF PRIMARY CLINICAL STUDIES

The applicant performed a clinical study in the United States and Japan to establish a reasonable assurance of safety and effectiveness of the Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System for treatment of *de novo* lesions in native coronary arteries that allow the use of a 2.0 mm diameter stent under IDE G140178. Data from this clinical study, in conjunction with data generated in the Global RESOLUTE Clinical Trial Program were the basis for the PMA Supplement approval decision. A summary of the clinical study is presented below.

### A. Study Design

Patients were treated between April 7, 2015 and February 23, 2016. The database for this PMA reflected data collected through March 21, 2017 and included 101 patients. There were 20 enrolling investigational sites across the United States and Japan.

The Medtronic RESOLUTE ONYX 2.0 mm Clinical Study was a single arm, open label, multi-center trial enrolling at least 100 subjects with ischemic heart disease attributable to stenotic lesions of the native coronary arteries that are amenable to percutaneous treatment with stenting. Subjects were to receive treatment of one or two lesions with stent diameters of 2.0 mm, one lesion per target vessel, for a maximum of two target vessels. Only one lesion was to be treated in a single target vessel. All treatment with the study stents was to be performed during a single index procedure.

If a subject became unstable before a study stent was attempted (stent introduced into guide catheter), the subject was not to be enrolled, or followed, and would not be included in the primary analysis of this trial. Another subject was to be enrolled to replace this subject for the primary analysis. If treatment with the Resolute Onyx stent was attempted (stent introduced into guide catheter) but not implanted, the subject would be considered part of the Intention-to-Treat population (ITT), followed through

the 12-month endpoint, and included in the primary analysis of this trial. After the 12-month follow up, these subjects would exit the study.

1. Clinical Inclusion and Exclusion Criteria

   Enrollment in the RESOLUTE ONYX 2.0 mm Clinical Study study was limited to patients who met the following Key Inclusion Criteria:

   - Subject has clinical evidence of ischemic heart disease. For single vessel disease: stable or unstable angina alone is sufficient. For multi-vessel treatment: a positive functional study or FFR to demonstrate functional need in 2.0 mm small vessel.
   - Subject has either a single target lesion suitable for treatment with a Resolute Onyx 2.0 mm stent, or two target lesions located in separate target vessels with at least one suitable for treatment with a Resolute Onyx 2.0 mm stent

   Patients were not permitted to be enrolled in the RESOLUTE ONYX 2.0 mm Clinical Study if they met any of the following Key Exclusion Criteria:

   - Evidence of an acute MI within 72 hours of the trial procedure
   - Planned PCI of any vessel within 30 days post-index procedure and/or planned PCI of the target vessel(s) within 12 months post-procedure. (No staged procedures)
   - Known hypersensitivity or contraindication to aspirin, heparin and bivalirudin, thienopyridines, cobalt, nickel, platinum, iridium, chromium, molybdenum, polymer coatings (e.g. BioLinx) or a sensitivity to contrast media, which cannot be adequately pre-medicated
   - History of an allergic reaction or significant sensitivity to drugs such as zotarolimus, rapamycin, tacrolimus, everolimus, or any other analogue or derivative
   - History of a stroke or transient ischemic attack (TIA) within the prior 6 months
   - Active peptic ulcer or upper gastrointestinal (GI) bleeding within the prior 6 months
   - History of bleeding diathesis or coagulopathy or will refuse blood transfusions
   - Concurrent medical condition with a life expectancy of less than 12 months
   - Currently participating in an investigational drug or another device trial that has not completed the primary endpoint
   - Documented left ventricular ejection fraction (LVEF) < 30% at the most recent evaluation

2. Follow-up Schedule

All patients were scheduled for follow up contacts at 30 days (± 5 days), 6 months (± 14 days), 12 months (± 30 days) and then annually (at 2 and 3 years ± 30 days) postoperatively. Patients in the 2.0 mm Angio subset were to return for angiographic assessment at 13 months (± 14 days). The key timepoints are shown in **Table 3.**

**Table 3: Schedule of Treatments and Assessments**

| Event | Index Hospitalization | | | | | Follow-up Assessments | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Screening/Pre-Procedure | | | Proced. | Post-Proced.[1] | 30 Day | 6 Month | 12 Month | 13 Month | 24, 36 Months |
| | Screen | Prior to procedure within: | | | | Subject Contact[2] | Subject Contact[2] or Clinic Visit[5] | Subject Contact[2] | Clinic Visit | Subject Contact[2] |
| | | 7 days | 72 hours | | | | | | | |
| Informed Consent signed | X | | | | | | | | | |
| Medical and cardiac history | X | | | | | | | | | |
| Angina status | X | | | | X | X | X | X | X | X |
| Pregnancy test[3] | | X | | | | | | | | |
| Creatinine | | X | | | | | | | | |
| WBC with Plts | | X | | | | | | | | |
| 12 lead ECG | | X | | | X Within 24 hours | | | | | |
| CK[6] and Troponin | | | X | | X 1st: >3hrs 2nd: >4 hrs after 1st, < 24 hrs | | | | | |
| MPI[5] | | | | | | | X | | | |
| QCA | | | | X | | | | | X[4] | |
| AE monitoring | | | | X | X | X | X | X | X | X SAEs only |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Antiplatelet medications | X Within 24 hours | | | | X | X | X | X | X | X |

1. End of procedure is defined as removal of the guide catheter
2. Subject contact includes phone call, email or clinic visit
3. For women of childbearing potential only
4. 2.0 mm Angio Subset only
5. Clinical visit for MPI assessment required for subjects that received an MPI at baseline and agree to do a follow-up MPI
6. If it is not standard hospital practice to measure CK values, CK-MB values are sufficient. All subjects should have Troponin values.

3. Clinical Endpoints

**Primary Endpoint(s)**

Target Lesion Failure (TLF) at 12-months post-procedure, defined as Cardiac Death, Target Vessel Myocardial Infarction (TVMI) (Q wave or non-Q wave) or Target Lesion Revascularization by percutaneous or surgical methods.

**Key Secondary Endpoint(s)**

Secondary endpoints of the trial include the following:

**Key Secondary Clinical Endpoints**

- Acute Success (Device, Lesion, Procedure)
- Cardiac Death
- Target Vessel Myocardial Infarction (TVMI)
- Cardiac Death and TVMI
- Major Adverse Cardiac Event (MACE)
- Target Lesion Failure (TLF)
- Target Vessel Failure (TVF)
- Stent Thrombosis (ST)

**Secondary Imaging Endpoint**

- Ischemia assessment per Myocardial Perfusion Imaging (MPI) for subjects who had MPI at baseline and consent to a follow up MPI

**Secondary Angiographic Endpoints**

- Late Lumen Loss (LLL)
- Binary Angiographic Restenosis (BAR) rate (defined as ≥ 50% diameter stenosis (DS))
- Percent Diameter Stenosis (% DS)

## B. Accountability of PMA Cohort

At the time of the database lock for this study, 101 subjects were eligible for the 12 month post-procedure follow up. **Figure 3** provides an overview of the subject accountability for this study through the 12-month Follow-Up visit. **Figure 4** provides an overview of the subject accountability for the Angiographic Subset at 13 months.

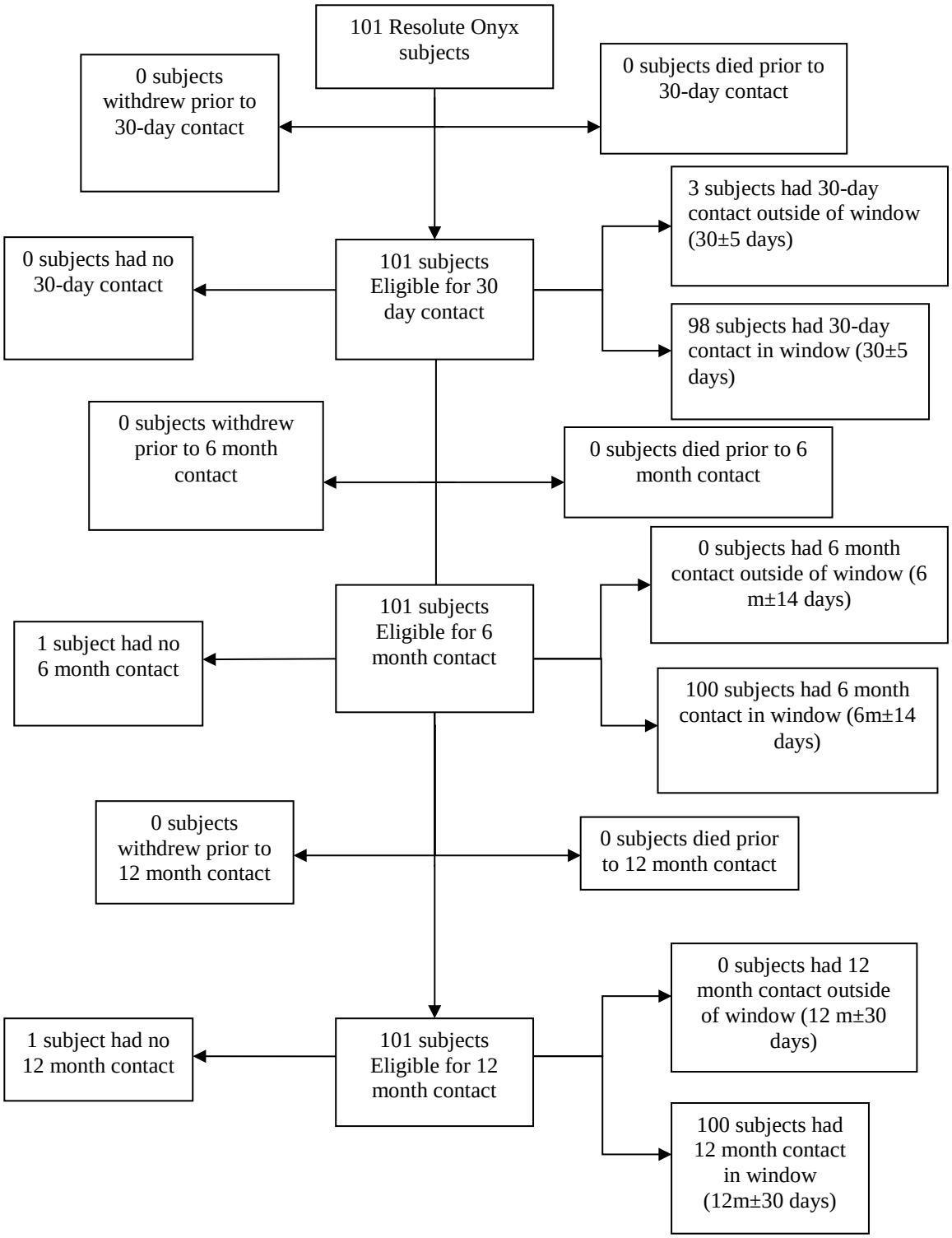



**Figure 3: RESOLUTE ONYX 2.0 mm Clinical Study Subject Accountability at 12- months Post-Procedure**

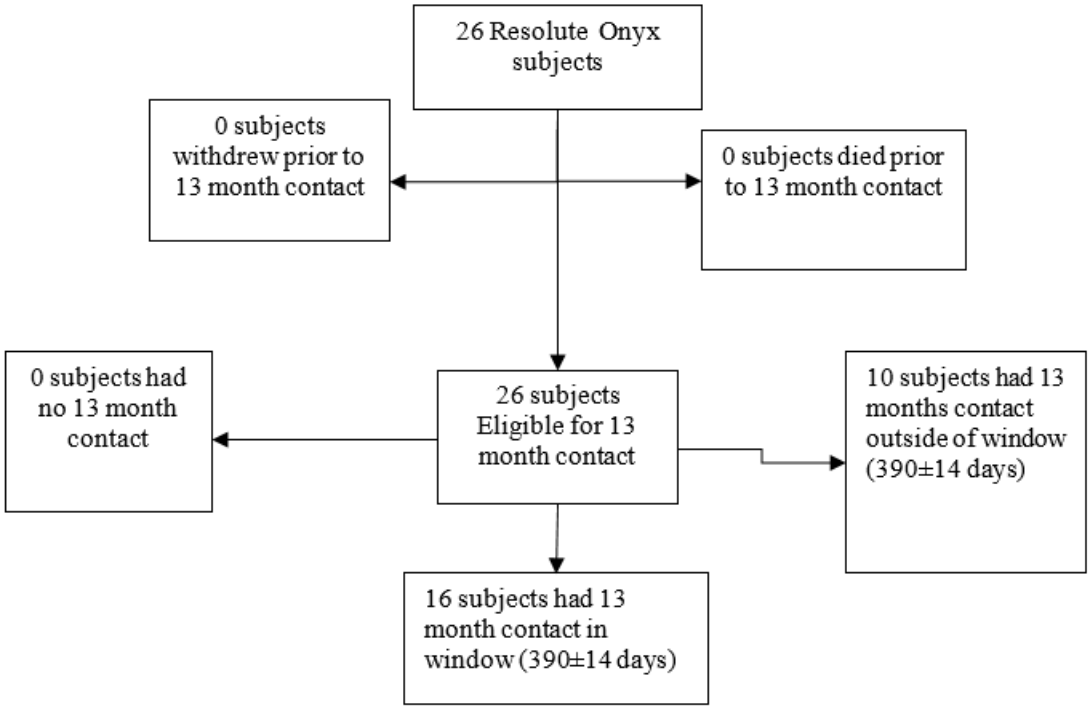



**Figure 4: RESOLUTE ONYX 2.0 mm Clinical Study Subject Accountability (Angiographic Subset at 13-months)**

## C. Study Population Demographics and Baseline Parameters

The demographics of the study population are typical for a PCI study performed in the US.

The mean age of the study subjects was 67.3 years, with 70.3% (71/101) of subjects being male, 46.5% (47/101) diabetics, 11.9% (12/101) were current smokers, 35.7% (35/98) had prior MI, 59.4% (60/101) had prior PCI, 82.2% (83/101) had hypertension, and 94.1% (95/101) reported hyperlipidemia. Baseline lesion characteristics include 36.6% (37/101) of subjects with LAD lesions, a mean lesion length of 12.59 ± 6.27mm, and 65.4% (68/104) ACC/AHA type B2/C lesions. The mean RVD was 1.91 ± 0.26 mm and the percentage diameter stenosis was 65.83 ± 10.89%.

## D. Safety and Effectiveness Results

### 1. Safety Results

The analysis of safety was based on the Primary cohort of 100 subjects available for the 12-month evaluation. There were no unanticipated adverse device effects (UADE), nor any device failures or malfunctions reported through 12 months. Adverse effects are reported in **Table 4** to **Table 7**.

**Adverse effects that occurred in the PMA clinical study:**

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| **Any Adverse Event** | 84.2% (85/101) |
| **BLOOD AND LYMPHATIC SYSTEM DISORDERS** | 5.0% (5/101) |
| ANAEMIA | 4.0% (4/101) |
| LEUKOCYTOSIS | 2.0% (2/101) |
| **CARDIAC DISORDERS** | 23.8% (24/101) |
| ACCELERATED IDIOVENTRICULAR RHYTHM | 1.0% (1/101) |
| ACUTE MYOCARDIAL INFARCTION | 1.0% (1/101) |
| ANGINA PECTORIS | 2.0% (2/101) |
| ARRHYTHMIA | 1.0% (1/101) |
| ATRIAL FIBRILLATION | 2.0% (2/101) |
| ATRIAL FLUTTER | 2.0% (2/101) |
| ATRIOVENTRICULAR BLOCK FIRST DEGREE | 1.0% (1/101) |
| BUNDLE BRANCH BLOCK LEFT | 1.0% (1/101) |
| CARDIAC ANEURYSM | 1.0% (1/101) |
| CARDIAC FAILURE CONGESTIVE | 2.0% (2/101) |
| CARDIAC FLUTTER | 1.0% (1/101) |
| CORONARY ARTERY DISEASE | 3.0% (3/101) |
| CORONARY ARTERY STENOSIS | 2.0% (2/101) |
| DIASTOLIC DYSFUNCTION | 1.0% (1/101) |
| IN-STENT CORONARY ARTERY RESTENOSIS | 1.0% (1/101) |
| MYOCARDIAL INFARCTION | 3.0% (3/101) |
| PALPITATIONS | 2.0% (2/101) |
| SINUS ARRHYTHMIA | 1.0% (1/101) |
| SINUS BRADYCARDIA | 1.0% (1/101) |
| SINUS TACHYCARDIA | 1.0% (1/101) |
| VENTRICULAR EXTRASYSTOLES | 1.0% (1/101) |
| VENTRICULAR TACHYCARDIA | 1.0% (1/101) |
| **EAR AND LABYRINTH DISORDERS** | 3.0% (3/101) |
| VERTIGO | 2.0% (2/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| VERTIGO POSITIONAL | 1.0% (1/101) |
| **ENDOCRINE DISORDERS** | 1.0% (1/101) |
| ADRENAL INSUFFICIENCY | 1.0% (1/101) |
| **EYE DISORDERS** | 5.0% (5/101) |
| CATARACT | 1.0% (1/101) |
| DIABETIC RETINAL OEDEMA | 1.0% (1/101) |
| EYE IRRITATION | 1.0% (1/101) |
| EYE SWELLING | 1.0% (1/101) |
| VISUAL ACUITY REDUCED | 1.0% (1/101) |
| **GASTROINTESTINAL DISORDERS** | 22.8% (23/101) |
| ABDOMINAL DISCOMFORT | 1.0% (1/101) |
| ABDOMINAL PAIN | 2.0% (2/101) |
| CONSTIPATION | 4.0% (4/101) |
| DIARRHOEA | 2.0% (2/101) |
| DYSPEPSIA | 2.0% (2/101) |
| DYSPHAGIA | 2.0% (2/101) |
| GASTRIC ULCER | 2.0% (2/101) |
| GASTROINTESTINAL HAEMORRHAGE | 2.0% (2/101) |
| HAEMATOCHEZIA | 1.0% (1/101) |
| HAEMORRHOIDAL HAEMORRHAGE | 1.0% (1/101) |
| HAEMORRHOIDS | 1.0% (1/101) |
| INGUINAL HERNIA | 1.0% (1/101) |
| IRRITABLE BOWEL SYNDROME | 1.0% (1/101) |
| NAUSEA | 5.9% (6/101) |
| PANCREATITIS | 1.0% (1/101) |
| PANCREATITIS ACUTE | 1.0% (1/101) |
| PERIODONTAL DISEASE | 1.0% (1/101) |
| PROCTALGIA | 1.0% (1/101) |
| TOOTHACHE | 1.0% (1/101) |
| VOMITING | 3.0% (3/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| **GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS** | 33.7% (34/101) |
| ASTHENIA | 2.0% (2/101) |
| CHEST DISCOMFORT | 3.0% (3/101) |
| CHEST PAIN | 15.8% (16/101) |
| CHILLS | 2.0% (2/101) |
| CYST | 1.0% (1/101) |
| FATIGUE | 4.0% (4/101) |
| GAIT DISTURBANCE | 1.0% (1/101) |
| INJECTION SITE BRUISING | 1.0% (1/101) |
| NON-CARDIAC CHEST PAIN | 5.0% (5/101) |
| OEDEMA | 1.0% (1/101) |
| OEDEMA PERIPHERAL | 1.0% (1/101) |
| PAIN | 2.0% (2/101) |
| PUNCTURE SITE HAEMORRHAGE | 3.0% (3/101) |
| PUNCTURE SITE PAIN | 1.0% (1/101) |
| PUNCTURE SITE REACTION | 1.0% (1/101) |
| PYREXIA | 1.0% (1/101) |
| VESSEL PUNCTURE SITE HAEMATOMA | 1.0% (1/101) |
| VESSEL PUNCTURE SITE HAEMORRHAGE | 2.0% (2/101) |
| **HEPATOBILIARY DISORDERS** | 2.0% (2/101) |
| BILIARY COLIC | 1.0% (1/101) |
| CHOLECYSTITIS ACUTE | 1.0% (1/101) |
| HEPATIC MASS | 1.0% (1/101) |
| **IMMUNE SYSTEM DISORDERS** | 3.0% (3/101) |
| HYPERSENSITIVITY | 2.0% (2/101) |
| SEASONAL ALLERGY | 1.0% (1/101) |
| **INFECTIONS AND INFESTATIONS** | 20.8% (21/101) |
| ABSCESS JAW | 1.0% (1/101) |
| ACUTE SINUSITIS | 1.0% (1/101) |
| BRONCHITIS | 4.0% (4/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| BRONCHITIS BACTERIAL | 1.0% (1/101) |
| CANDIDIASIS | 1.0% (1/101) |
| CLOSTRIDIUM DIFFICILE COLITIS | 1.0% (1/101) |
| DIVERTICULITIS | 1.0% (1/101) |
| HERPES ZOSTER | 1.0% (1/101) |
| IMPETIGO | 1.0% (1/101) |
| KERATITIS HERPETIC | 1.0% (1/101) |
| LARYNGITIS | 1.0% (1/101) |
| PHARYNGITIS | 1.0% (1/101) |
| PNEUMONIA | 2.0% (2/101) |
| PNEUMONIA MYCOPLASMAL | 1.0% (1/101) |
| POST PROCEDURAL INFECTION | 1.0% (1/101) |
| SEPSIS | 1.0% (1/101) |
| STAPHYLOCOCCAL SEPSIS | 1.0% (1/101) |
| UPPER RESPIRATORY TRACT INFECTION | 2.0% (2/101) |
| URINARY TRACT INFECTION | 4.0% (4/101) |
| **INJURY, POISONING AND PROCEDURAL COMPLICATIONS** | 13.9% (14/101) |
| ANAEMIA POSTOPERATIVE | 1.0% (1/101) |
| ANKLE FRACTURE | 1.0% (1/101) |
| CONTUSION | 4.0% (4/101) |
| FALL | 3.0% (3/101) |
| IN-STENT ARTERIAL RESTENOSIS | 2.0% (2/101) |
| JOINT INJURY | 1.0% (1/101) |
| POST PROCEDURAL HAEMATOMA | 1.0% (1/101) |
| POST PROCEDURAL HAEMORRHAGE | 1.0% (1/101) |
| PROCEDURAL HYPOTENSION | 1.0% (1/101) |
| ROAD TRAFFIC ACCIDENT | 1.0% (1/101) |
| TRAUMATIC HAEMATOMA | 1.0% (1/101) |
| **INVESTIGATIONS** | 26.7% (27/101) |
| BLOOD PRESSURE INCREASED | 1.0% (1/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| BLOOD THYROID STIMULATING HORMONE DECREASED | 1.0% (1/101) |
| CARDIAC ENZYMES INCREASED | 11.9% (12/101) |
| CAROTID BRUIT | 1.0% (1/101) |
| ELECTROCARDIOGRAM CHANGE | 1.0% (1/101) |
| FREE PROSTATE-SPECIFIC ANTIGEN DECREASED | 1.0% (1/101) |
| HAEMATOCRIT DECREASED | 1.0% (1/101) |
| HAEMOGLOBIN DECREASED | 1.0% (1/101) |
| HEART RATE IRREGULAR | 1.0% (1/101) |
| TROPONIN I INCREASED | 5.0% (5/101) |
| TROPONIN INCREASED | 5.9% (6/101) |
| WEIGHT DECREASED | 1.0% (1/101) |
| **METABOLISM AND NUTRITION DISORDERS** | 11.9% (12/101) |
| DIABETES MELLITUS | 2.0% (2/101) |
| DIABETES MELLITUS NON-INSULIN-DEPENDENT | 2.0% (2/101) |
| ELECTROLYTE IMBALANCE | 1.0% (1/101) |
| GOUT | 1.0% (1/101) |
| HYPERGLYCAEMIA | 2.0% (2/101) |
| HYPERKALAEMIA | 1.0% (1/101) |
| HYPOGLYCAEMIA | 2.0% (2/101) |
| HYPOPHOSPHATAEMIA | 1.0% (1/101) |
| METABOLIC ACIDOSIS | 1.0% (1/101) |
| **MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS** | 22.8% (23/101) |
| ARTHRALGIA | 2.0% (2/101) |
| BACK PAIN | 7.9% (8/101) |
| COSTOCHONDRITIS | 1.0% (1/101) |
| JOINT SWELLING | 1.0% (1/101) |
| LIMB DISCOMFORT | 1.0% (1/101) |
| MUSCLE SPASMS | 4.0% (4/101) |
| MUSCULAR WEAKNESS | 1.0% (1/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| MUSCULOSKELETAL DISCOMFORT | 2.0% (2/101) |
| MUSCULOSKELETAL PAIN | 3.0% (3/101) |
| OSTEOARTHROPATHY | 1.0% (1/101) |
| PAIN IN EXTREMITY | 4.0% (4/101) |
| ROTATOR CUFF SYNDROME | 1.0% (1/101) |
| **NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS)** | 7.9% (8/101) |
| BASAL CELL CARCINOMA | 1.0% (1/101) |
| ESSENTIAL THROMBOCYTHAEMIA | 1.0% (1/101) |
| LIP AND/OR ORAL CAVITY CANCER | 1.0% (1/101) |
| LUNG NEOPLASM | 1.0% (1/101) |
| MYELODYSPLASTIC SYNDROME | 1.0% (1/101) |
| OROPHARYNGEAL CANCER RECURRENT | 1.0% (1/101) |
| SEBORRHOEIC KERATOSIS | 1.0% (1/101) |
| SQUAMOUS CELL CARCINOMA | 1.0% (1/101) |
| TRANSITIONAL CELL CARCINOMA | 1.0% (1/101) |
| **NERVOUS SYSTEM DISORDERS** | 19.8% (20/101) |
| AGEUSIA | 1.0% (1/101) |
| CAROTID ARTERY STENOSIS | 2.0% (2/101) |
| CEREBROVASCULAR ACCIDENT | 2.0% (2/101) |
| CLUSTER HEADACHE | 1.0% (1/101) |
| DEMENTIA | 1.0% (1/101) |
| DIZZINESS | 4.0% (4/101) |
| ENCEPHALOPATHY | 1.0% (1/101) |
| HEADACHE | 5.0% (5/101) |
| HYPOAESTHESIA | 2.0% (2/101) |
| NEUROPATHY | 1.0% (1/101) |
| SOMNOLENCE | 1.0% (1/101) |
| SYNCOPE | 3.0% (3/101) |
| SYNCOPE VASOVAGAL | 1.0% (1/101) |
| **PSYCHIATRIC DISORDERS** | 3.0% (3/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| ANXIETY | 1.0% (1/101) |
| DEPRESSION | 2.0% (2/101) |
| **RENAL AND URINARY DISORDERS** | 8.9% (9/101) |
| BLADDER OBSTRUCTION | 1.0% (1/101) |
| HAEMATURIA | 4.0% (4/101) |
| NEPHROLITHIASIS | 2.0% (2/101) |
| POLLAKIURIA | 1.0% (1/101) |
| RENAL FAILURE ACUTE | 2.0% (2/101) |
| RENAL FAILURE CHRONIC | 1.0% (1/101) |
| URINARY RETENTION | 1.0% (1/101) |
| **REPRODUCTIVE SYSTEM AND BREAST DISORDERS** | 2.0% (2/101) |
| BENIGN PROSTATIC HYPERPLASIA | 1.0% (1/101) |
| ERECTILE DYSFUNCTION | 1.0% (1/101) |
| **RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS** | 27.7% (28/101) |
| ACUTE RESPIRATORY DISTRESS SYNDROME | 1.0% (1/101) |
| ACUTE RESPIRATORY FAILURE | 1.0% (1/101) |
| CHRONIC OBSTRUCTIVE PULMONARY DISEASE | 2.0% (2/101) |
| COUGH | 2.0% (2/101) |
| DYSPNOEA | 16.8% (17/101) |
| DYSPNOEA EXERTIONAL | 2.0% (2/101) |
| EPISTAXIS | 4.0% (4/101) |
| NASAL CONGESTION | 1.0% (1/101) |
| PHARYNGOLARYNGEAL PAIN | 1.0% (1/101) |
| RESPIRATORY ACIDOSIS | 1.0% (1/101) |
| RESPIRATORY FAILURE | 1.0% (1/101) |
| SLEEP APNOEA SYNDROME | 1.0% (1/101) |
| **SKIN AND SUBCUTANEOUS TISSUE DISORDERS** | 5.0% (5/101) |
| ECZEMA | 1.0% (1/101) |
| HYPERHIDROSIS | 1.0% (1/101) |
| HYPOAESTHESIA FACIAL | 1.0% (1/101) |

**Table 4: All Site Reported Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| PRURITUS | 1.0% (1/101) |
| SKIN LESION | 1.0% (1/101) |
| **VASCULAR DISORDERS** | 18.8% (19/101) |
| AORTIC ANEURYSM | 1.0% (1/101) |
| ARTERIOSCLEROSIS | 1.0% (1/101) |
| ARTERIOSCLEROSIS OBLITERANS | 1.0% (1/101) |
| HAEMATOMA | 1.0% (1/101) |
| HAEMORRHAGE | 1.0% (1/101) |
| HYPERTENSION | 7.9% (8/101) |
| HYPOTENSION | 1.0% (1/101) |
| INTERMITTENT CLAUDICATION | 1.0% (1/101) |
| ORTHOSTATIC HYPOTENSION | 2.0% (2/101) |
| PERIPHERAL ARTERY ANEURYSM | 1.0% (1/101) |
| PERIPHERAL VASCULAR DISORDER | 2.0% (2/101) |
| VARICOSE VEIN | 2.0% (2/101) |

[1] Numerator (m) is the number of subjects with the specific classification, denominator (n) is the number of subjects in the study
group with known values, and percentage (%) was calculated as 100 × (m/n)

**Table 5: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| **Any Serious Adverse Events** | 37.6% (38/101) |
| **BLOOD AND LYMPHATIC SYSTEM DISORDERS** | 3.0% (3/101) |
| ANAEMIA | 3.0% (3/101) |
| **CARDIAC DISORDERS** | 10.9% (11/101) |
| ACUTE MYOCARDIAL INFARCTION | 1.0% (1/101) |
| ANGINA PECTORIS | 1.0% (1/101) |
| ARRHYTHMIA | 1.0% (1/101) |

**Table 5: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| ATRIAL FLUTTER | 2.0% (2/101) |
| CARDIAC ANEURYSM | 1.0% (1/101) |
| CARDIAC FAILURE CONGESTIVE | 1.0% (1/101) |
| CORONARY ARTERY DISEASE | 3.0% (3/101) |
| CORONARY ARTERY STENOSIS | 2.0% (2/101) |
| IN-STENT CORONARY ARTERY RESTENOSIS | 1.0% (1/101) |
| VENTRICULAR TACHYCARDIA | 1.0% (1/101) |
| **EAR AND LABYRINTH DISORDERS** | 1.0% (1/101) |
| VERTIGO POSITIONAL | 1.0% (1/101) |
| **ENDOCRINE DISORDERS** | 1.0% (1/101) |
| ADRENAL INSUFFICIENCY | 1.0% (1/101) |
| **EYE DISORDERS** | 1.0% (1/101) |
| EYE SWELLING | 1.0% (1/101) |
| **GASTROINTESTINAL DISORDERS** | 7.9% (8/101) |
| GASTRIC ULCER | 2.0% (2/101) |
| GASTROINTESTINAL HAEMORRHAGE | 2.0% (2/101) |
| INGUINAL HERNIA | 1.0% (1/101) |
| IRRITABLE BOWEL SYNDROME | 1.0% (1/101) |
| NAUSEA | 1.0% (1/101) |
| PANCREATITIS ACUTE | 1.0% (1/101) |
| **GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS** | 4.0% (4/101) |
| CHEST DISCOMFORT | 1.0% (1/101) |
| CHEST PAIN | 3.0% (3/101) |
| CHILLS | 1.0% (1/101) |
| PYREXIA | 1.0% (1/101) |
| **HEPATOBILIARY DISORDERS** | 1.0% (1/101) |
| BILIARY COLIC | 1.0% (1/101) |
| CHOLECYSTITIS ACUTE | 1.0% (1/101) |
| **INFECTIONS AND INFESTATIONS** | 8.9% (9/101) |
| BRONCHITIS | 1.0% (1/101) |
| BRONCHITIS BACTERIAL | 1.0% (1/101) |

**Table 5: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| CLOSTRIDIUM DIFFICILE COLITIS | 1.0% (1/101) |
| PNEUMONIA MYCOPLASMAL | 1.0% (1/101) |
| POST PROCEDURAL INFECTION | 1.0% (1/101) |
| STAPHYLOCOCCAL SEPSIS | 1.0% (1/101) |
| URINARY TRACT INFECTION | 3.0% (3/101) |
| **INJURY, POISONING AND PROCEDURAL COMPLICATIONS** | 4.0% (4/101) |
| FALL | 1.0% (1/101) |
| IN-STENT ARTERIAL RESTENOSIS | 2.0% (2/101) |
| JOINT INJURY | 1.0% (1/101) |
| **INVESTIGATIONS** | 3.0% (3/101) |
| CARDIAC ENZYMES INCREASED | 1.0% (1/101) |
| HAEMATOCRIT DECREASED | 1.0% (1/101) |
| HAEMOGLOBIN DECREASED | 1.0% (1/101) |
| HEART RATE IRREGULAR | 1.0% (1/101) |
| **METABOLISM AND NUTRITION DISORDERS** | 2.0% (2/101) |
| DIABETES MELLITUS NON-INSULIN-DEPENDENT | 1.0% (1/101) |
| HYPERGLYCAEMIA | 1.0% (1/101) |
| **MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS** | 2.0% (2/101) |
| MUSCULOSKELETAL PAIN | 1.0% (1/101) |
| ROTATOR CUFF SYNDROME | 1.0% (1/101) |
| **NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS)** | 3.0% (3/101) |
| LIP AND/OR ORAL CAVITY CANCER | 1.0% (1/101) |
| OROPHARYNGEAL CANCER RECURRENT | 1.0% (1/101) |
| TRANSITIONAL CELL CARCINOMA | 1.0% (1/101) |
| **NERVOUS SYSTEM DISORDERS** | 7.9% (8/101) |
| CAROTID ARTERY STENOSIS | 2.0% (2/101) |
| CEREBROVASCULAR ACCIDENT | 2.0% (2/101) |
| ENCEPHALOPATHY | 1.0% (1/101) |
| SYNCOPE | 2.0% (2/101) |

**Table 5: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| SYNCOPE VASOVAGAL | 1.0% (1/101) |
| **RENAL AND URINARY DISORDERS** | 2.0% (2/101) |
| HAEMATURIA | 1.0% (1/101) |
| NEPHROLITHIASIS | 1.0% (1/101) |
| **RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS** | 3.0% (3/101) |
| ACUTE RESPIRATORY DISTRESS SYNDROME | 1.0% (1/101) |
| CHRONIC OBSTRUCTIVE PULMONARY DISEASE | 1.0% (1/101) |
| DYSPNOEA | 1.0% (1/101) |
| RESPIRATORY FAILURE | 1.0% (1/101) |
| **VASCULAR DISORDERS** | 5.9% (6/101) |
| AORTIC ANEURYSM | 1.0% (1/101) |
| ARTERIOSCLEROSIS | 1.0% (1/101) |
| ARTERIOSCLEROSIS OBLITERANS | 1.0% (1/101) |
| INTERMITTENT CLAUDICATION | 1.0% (1/101) |
| ORTHOSTATIC HYPOTENSION | 1.0% (1/101) |
| PERIPHERAL VASCULAR DISORDER | 1.0% (1/101) |

[1] Numerator (m) is the number of subjects with the specific classification, denominator (n) is the number of subjects in the
studygroup with known values, and percentage (%) was calculated as 100 × (m/n)

**Table 6: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| **Any Serious Adverse Events** | 37.6% (38/101) |
| **BLOOD AND LYMPHATIC SYSTEM DISORDERS** | 3.0% (3/101) |
| ANAEMIA | 3.0% (3/101) |
| **CARDIAC DISORDERS** | 10.9% (11/101) |
| ACUTE MYOCARDIAL INFARCTION | 1.0% (1/101) |
| ANGINA PECTORIS | 1.0% (1/101) |

**Table 6: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| ARRHYTHMIA | 1.0% (1/101) |
| ATRIAL FLUTTER | 2.0% (2/101) |
| CARDIAC ANEURYSM | 1.0% (1/101) |
| CARDIAC FAILURE CONGESTIVE | 1.0% (1/101) |
| CORONARY ARTERY DISEASE | 3.0% (3/101) |
| CORONARY ARTERY STENOSIS | 2.0% (2/101) |
| IN-STENT CORONARY ARTERY RESTENOSIS | 1.0% (1/101) |
| VENTRICULAR TACHYCARDIA | 1.0% (1/101) |
| **EAR AND LABYRINTH DISORDERS** | 1.0% (1/101) |
| VERTIGO POSITIONAL | 1.0% (1/101) |
| **ENDOCRINE DISORDERS** | 1.0% (1/101) |
| ADRENAL INSUFFICIENCY | 1.0% (1/101) |
| **EYE DISORDERS** | 1.0% (1/101) |
| EYE SWELLING | 1.0% (1/101) |
| **GASTROINTESTINAL DISORDERS** | 7.9% (8/101) |
| GASTRIC ULCER | 2.0% (2/101) |
| GASTROINTESTINAL HAEMORRHAGE | 2.0% (2/101) |
| INGUINAL HERNIA | 1.0% (1/101) |
| IRRITABLE BOWEL SYNDROME | 1.0% (1/101) |
| NAUSEA | 1.0% (1/101) |
| PANCREATITIS ACUTE | 1.0% (1/101) |
| **GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS** | 4.0% (4/101) |
| CHEST DISCOMFORT | 1.0% (1/101) |
| CHEST PAIN | 3.0% (3/101) |
| CHILLS | 1.0% (1/101) |
| PYREXIA | 1.0% (1/101) |
| **HEPATOBILIARY DISORDERS** | 1.0% (1/101) |
| BILIARY COLIC | 1.0% (1/101) |
| CHOLECYSTITIS ACUTE | 1.0% (1/101) |
| **INFECTIONS AND INFESTATIONS** | 8.9% (9/101) |

**Table 6: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| BRONCHITIS | 1.0% (1/101) |
| BRONCHITIS BACTERIAL | 1.0% (1/101) |
| CLOSTRIDIUM DIFFICILE COLITIS | 1.0% (1/101) |
| PNEUMONIA MYCOPLASMAL | 1.0% (1/101) |
| POST PROCEDURAL INFECTION | 1.0% (1/101) |
| STAPHYLOCOCCAL SEPSIS | 1.0% (1/101) |
| URINARY TRACT INFECTION | 3.0% (3/101) |
| **INJURY, POISONING AND PROCEDURAL COMPLICATIONS** | 4.0% (4/101) |
| FALL | 1.0% (1/101) |
| IN-STENT ARTERIAL RESTENOSIS | 2.0% (2/101) |
| JOINT INJURY | 1.0% (1/101) |
| **INVESTIGATIONS** | 3.0% (3/101) |
| CARDIAC ENZYMES INCREASED | 1.0% (1/101) |
| HAEMATOCRIT DECREASED | 1.0% (1/101) |
| HAEMOGLOBIN DECREASED | 1.0% (1/101) |
| HEART RATE IRREGULAR | 1.0% (1/101) |
| **METABOLISM AND NUTRITION DISORDERS** | 2.0% (2/101) |
| DIABETES MELLITUS NON-INSULIN-DEPENDENT | 1.0% (1/101) |
| HYPERGLYCAEMIA | 1.0% (1/101) |
| **MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS** | 2.0% (2/101) |
| MUSCULOSKELETAL PAIN | 1.0% (1/101) |
| ROTATOR CUFF SYNDROME | 1.0% (1/101) |
| **NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS)** | 3.0% (3/101) |
| LIP AND/OR ORAL CAVITY CANCER | 1.0% (1/101) |
| OROPHARYNGEAL CANCER RECURRENT | 1.0% (1/101) |
| TRANSITIONAL CELL CARCINOMA | 1.0% (1/101) |
| **NERVOUS SYSTEM DISORDERS** | 7.9% (8/101) |
| CAROTID ARTERY STENOSIS | 2.0% (2/101) |

**Table 6: All Site Reported Serious Adverse Events by Type**

| Events | RESOLUTE ONYX 2.0mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| CEREBROVASCULAR ACCIDENT | 2.0% (2/101) |
| ENCEPHALOPATHY | 1.0% (1/101) |
| SYNCOPE | 2.0% (2/101) |
| SYNCOPE VASOVAGAL | 1.0% (1/101) |
| **RENAL AND URINARY DISORDERS** | 2.0% (2/101) |
| HAEMATURIA | 1.0% (1/101) |
| NEPHROLITHIASIS | 1.0% (1/101) |
| **RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS** | 3.0% (3/101) |
| ACUTE RESPIRATORY DISTRESS SYNDROME | 1.0% (1/101) |
| CHRONIC OBSTRUCTIVE PULMONARY DISEASE | 1.0% (1/101) |
| DYSPNOEA | 1.0% (1/101) |
| RESPIRATORY FAILURE | 1.0% (1/101) |
| **VASCULAR DISORDERS** | 5.9% (6/101) |
| AORTIC ANEURYSM | 1.0% (1/101) |
| ARTERIOSCLEROSIS | 1.0% (1/101) |
| ARTERIOSCLEROSIS OBLITERANS | 1.0% (1/101) |
| INTERMITTENT CLAUDICATION | 1.0% (1/101) |
| ORTHOSTATIC HYPOTENSION | 1.0% (1/101) |
| PERIPHERAL VASCULAR DISORDER | 1.0% (1/101) |

[1] Numerator (m) is the number of subjects with the specific classification, denominator (n) is the number of subjects in the study group with known values, and percentage (%) was calculated as 100 x (m/n)

**Table 7: Stent Thrombosis (ARC Definition)**

| Stent Thrombosis to 240 days | RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1] |
|---|---|
| Stent Thrombosis Related to Non Target Lesions to 360 Days | 0.0% (0/100) |

**Table 7: Stent Thrombosis (ARC Definition)**

| **Stent Thrombosis to 240 days** | **RESOLUTE ONYX 2.0 mm (N=101 Subjects) %(m/n)[1]** |
|---|---|
| Early (0 to 30 Days) | 0.0% (0/100) |
| Late (31 to 360 Days) | 0.0% (0/100) |
| Overall Stent Thrombosis to 360 Days | |
| Definite ST | 0.0% (0/100) |
| Probable ST | 0.0% (0/100) |
| Definite + Probable ST | 0.0% (0/100) |
| Early (0 to 30 Days) | |
| Definite ST | 0.0% (0/100) |
| Probable ST | 0.0% (0/100) |
| Definite + Probable ST | 0.0% (0/100) |
| Late (31 to 360 Days) | |
| Definite ST | 0.0% (0/100) |
| Probable ST | 0.0% (0/100) |
| Definite + Probable ST | 0.0% (0/100) |
| [1]Numerator (m) is the number of subjects with the specific classification, denominator (n) is the number of subjects in the study group with known values, and percentage (%) was calculated as 100 × (m/n) | |

2. Effectiveness Results

   The analysis of effectiveness was based on the 100 evaluable patients at the 12-month time point. The primary endpoint of target lesion failure (TLF) at 12-months post-procedure demonstrated that the rate of TLF in the ITT primary analysis set at 12 months was 5.0% (5/100), fulfilling the pre-specified performance criterion (upper 1-sided 95% CI of 10.2%, compared with the performance goal of 19%, $p < 0.001$). The primary endpoint was also analyzed by gender, resulting in a TLF rate of 7.1% (5/70) in male subjects and 0.0% (0/30) in female subjects. Key outcomes for this study are presented below in **Table 8** to **Table 11**.

**Table 8: RESOLUTE ONYX 2.0 mm Clinical Study – Primary Endpoint Analysis**

| Primary Endpoint - TLF at 12-month | Resolute Onyx 2.0mm (N = 101 Subjects) | One-sided Upper 95% Confidence Interval[1] | Performance Goal |
|---|---|---|---|
| Primary Analysis – with analysis lesion only[2] | | | |
| – ITT set | 5.0% (5/100) | 10.2% | 19% |
| – PP set | 2.2% (2/90) | 6.8% | 19% |
| Secondary Analysis – with all lesions included[3] | | | |
| – ITT set | 5.0% (5/100) | 10.2% | 19% |
| – PP set | 2.2% (2/90) | 6.8% | 19% |
| Multiple Imputation[4] | | | |
| – ITT set | 5.0% | 8.6% | 19% |
| – PP set | 2.3% | 4.9% | 19% |
| Tipping Point Analysis[5] | | | |
| – ITT set | 5.9% (6/101) | 11.4% | 19% |
| – PP set | 3.3% (3/91) | 8.3% | 19% |
| Worst Case Analysis[6] | | | |
| – ITT set | 5.9% (6/101) | 11.4% | 19% |
| – PP set | 3.3% (3/91) | 8.3% | 19% |

[1] The one-sided upper 95% CI is calculated by binomial (exact) distribution
[2] The lesions with a Resolute Onyx 2.0mm stent are included in the analysis. For 2 or more lesions with Resolute Onyx 2.0mm stents per subject, the lesion is randomly selected.
[3] All target lesions are included in the analysis.
[4] The covariates to be used in the imputation model are lesion-length, baseline RVD, age, sex, diabetes, history of MI, Canadian Cardiovascular Society Angina Class, and TLF-missing status at visits prior to dropout. The longest lesion length
and the smallest baseline RVD are used for the subjects have 2 or more target lesions in the impu tation model.
[5] Impute the most 12-month TLF-missing status as yes so that the one-side upper 95% confidence interval of 12-month TLF rate can be less than or equal to the performance goal.
[6] Impute all the 12-month TLF-missing status as yes.

**Table 9: Primary Endpoint Analysis by Gender**

| Primary Endpoint | Male (N = 71 Subjects) | Female (N = 30 Subjects) |
|---|---|---|
| Target Lesion Failure to 12 months | 7.1% (5/70) | 0.0% (0/30) |

**Table 10: RESOLUTE ONYX 2.0 mm Clinical Study – Principal Safety and Effectiveness**

| Safety and Effectiveness Measures | RESOLUTE ONYX 2.0mm (N=101 Subjects N=104 Lesions) %(m/n)[1] |
|---|---|
| **In-Hospital** | |
| Target Lesion Failure (TLF)[2] | 2.0% (2/101) |
| Target Vessel Failure (TVF)[3] | 2.0% (2/101) |
| MACE[4] | 2.0% (2/101) |
| Cardiac Death or Target Vessel MI (TVMI) | 2.0% (2/101) |
| Death or TVMI | 2.0% (2/101) |
| Death | 0.0% (0/101) |
| Cardiac Death | 0.0% (0/101) |
| Non Cardiac Death | 0.0% (0/101) |
| TVMI (Extended historical definition) | 2.0% (2/101) |
| Clinically Driven TLR | 0.0% (0/101) |
| Clinically Driven TVR | 0.0% (0/101) |
| Stent Thrombosis (ARC) Definite/Probable[5] | 0.0% (0/101) |
| **Safety Measures (to 180 days)** | |
| Target Lesion Failure (TLF)[2] | 4.0% (4/100) |
| Target Vessel Failure (TVF)[3] | 4.0% (4/100) |
| MACE[4] | 4.0% (4/100) |
| Cardiac Death or Target Vessel MI (TVMI) | 3.0% (3/100) |
| Death or TVMI | 3.0% (3/100) |
| Death | 0.0% (0/100) |
| Cardiac Death | 0.0% (0/100) |
| Non Cardiac Death | 0.0% (0/100) |
| TVMI (Extended historical definition) | 3.0% (3/100) |

**Table 10: RESOLUTE ONYX 2.0 mm Clinical Study – Principal Safety and Effectiveness**

| **Safety and Effectiveness Measures** | **RESOLUTE ONYX 2.0mm (N=101 Subjects N=104 Lesions) %(m/n)[1]** |
|---|---|
| Clinically Driven TLR | 1.0% (1/100) |
| Clinically Driven TVR | 1.0% (1/100) |
| Stent Thrombosis (ARC) Definite/Probable | 0.0% (0/100) |
| **Safety Measures (to 360 days)** | |
| Target Lesion Failure (TLF)[2] | 5.0% (5/100) |
| Target Vessel Failure (TVF)[3] | 5.0% (5/100) |
| MACE[4] | 5.0% (5/100) |
| Cardiac Death or Target Vessel MI (TVMI) | 3.0% (3/100) |
| Death or TVMI | 3.0% (3/100) |
| Death | 0.0% (0/100) |
| Cardiac Death | 0.0% (0/100) |
| Non Cardiac Death | 0.0% (0/100) |
| TVMI (Extended historical definition) | 3.0% (3/100) |
| Clinically Driven TLR | 2.0% (2/100) |
| Clinically Driven TVR | 2.0% (2/100) |
| Stent Thrombosis (ARC) Definite/Probable | 0.0% (0/100) |
| Early Thrombosis (<=30 days) | 0.0% (0/100) |
| Late Thrombosis (31-360 days) | 0.0% (0/100) |
| **Angiography (13 months)** | |
| Percent Diameter Stenosis (% DS) | |
| In-stent | |
| n | 25 |
| Mean±SD | 22.49 ± 26.89 |
| Median (Q1, Q3) | 15.66 (9.57, 31.72) |
| Min, Max | -26.71, 100.00 |
| In-segment | |
| n | 25 |
| Mean±SD | 37.92 ± 21.54 |
| Median (Q1, Q3) | 31.72 (23.54, 42.50) |

**Table 10: RESOLUTE ONYX 2.0 mm Clinical Study – Principal Safety and Effectiveness**

| **Safety and Effectiveness Measures** | **RESOLUTE ONYX 2.0mm (N=101 Subjects N=104 Lesions) %(m/n)[1]** |
|---|---|
| Min, Max | 14.06, 100.00 |
| Minimal Lumen Diameter (mm) | |
| In-stent | |
| n | 25 |
| Mean±SD | 1.55 ± 0.52 |
| Median (Q1, Q3) | 1.63 (1.53, 1.81) |
| Min, Max | 0.00, 2.20 |
| In-segment | |
| n | 25 |
| Mean±SD | 1.25 ± 0.46 |
| Median (Q1, Q3) | 1.44 (1.09, 1.52) |
| Min, Max | 0.00, 1.77 |
| Late Luminal Loss (mm) | |
| In-stent | |
| n | 25 |
| Mean±SD | 0.26 ± 0.48 |
| Median (Q1, Q3) | 0.06 (0.00, 0.33) |
| Min, Max | -0.42, 1.58 |
| In-segment | |
| n | 25 |
| Mean±SD | 0.25 ± 0.41 |
| Median (Q1, Q3) | 0.21 (-0.08, 0.42) |
| Min, Max | -0.39, 1.30 |
| In-Stent Binary Angiographic Restenosis (BAR) Rate | 12.0% (3/25) |
| In-Segment Binary Angiographic Restenosis (BAR) Rate | 20.0% (5/25) |
| **Effectiveness Measures** | |
| Lesion Success[5] | 99.0% (103/104) |
| Device Success[6] | 96.2% (100/104) |

**Table 10: RESOLUTE ONYX 2.0 mm Clinical Study – Principal Safety and Effectiveness**

| Safety and Effectiveness Measures | RESOLUTE ONYX 2.0mm (N=101 Subjects N=104 Lesions) %(m/n)[1] |
|---|---|
| Procedure Success[7] | 97.0% (98/101) |

[1]Numerator (m) is the number of Subjects with the specific classification, denominator (n) is the number of Subjects in the study
group with known values, and percentage (%) was calculated as 100 × (m/n)
[2]Cardiac death, target vessel myocardial infarction (Q wave and non Q wave) or clinically-driven target lesion revascularization (TLR) by percutaneous or surgical methods.
[3]Cardiac death, target vessel myocardial infarction (Q wave and non Q wave) or clinically-driven target vessel revascularization (TVR) by percutaneous or surgical methods.
[4]Defined as death, myocardial infarction (Q wave and non Q-wave), emergent coronary bypass surgery, or repeat target lesion
revascularization (clinically driven/clinically indicated) by percutaneous or surgical methods.
[5]The attainment of < 30% residual stenosis by QCA (or < 20% by visual assessment) AND TIMI flow 3 after the procedure, using any percutaneous method.
[6]The attainment of < 30% residual stenosis by QCA (or < 20% by visual assessment) AND TIMI flow 3 after the procedure, using the assigned device only.
[7]The attainment of < 30% residual stenosis by QCA (or < 20% by visual assessment) AND TIMI flow 3 after the procedure, using
any percutaneous method without the occurrence of MACE during the hospital stay.
Extended historical definition of MI is used for all the composite endpoints.

**Table 11: RESOLUTE ONYX 2.0 mm Clinical Study – ARC Defined Definite/Probable Stent Thrombosis through 12 Months**

| | Resolute ONYX™ (N=101 Subjects N=104 Lesions) %(m/n)[1] |
|---|---|
| Stent Thrombosis | 0.0% (0/100) |
| Early Thrombosis (<=30 days) | 0.0% (0/100) |
| Late Thrombosis (31-360 days) | 0.0% (0/100) |

***Notes***
N = The total number of subjects enrolled.
Numbers are % (Count/Number of Eligible Subjects).
Subjects are only counted once for each time period.
12-month timeframe includes follow-up window (360 days ± 30 days).

3. Subgroup Analyses
   The RESOLUTE ONYX 2.0 mm Clinical Study did not include prespecified subgroup analyses.

4. Pediatric Extrapolation
   In this premarket application, existing clinical data was not leveraged to support approval of a pediatric patient population.

## XI. FINANCIAL DISCLOSURE

The Financial Disclosure by Clinical Investigators regulation (21 CFR 54) requires applicants who submit a marketing application to include certain information concerning the compensation to, and financial interests and arrangement of, any clinical investigator conducting clinical studies covered by the regulation. The RESOLUTE ONYX 2.0 mm Clinical Study included 187 investigators of which none were full-time or part-time employees of the sponsor and eight (8) had disclosable financial interests/arrangements as defined in 21 CFR 54.2(a), (b), (c) and (f) and described below:

- Compensation to the investigator for conducting the study where the value could be influenced by the outcome of the study: none
- Significant payment of other sorts: eight (8)
- Proprietary interest in the product tested held by the investigator: none
- Significant equity interest held by investigator in sponsor of covered study: none

The applicant has adequately disclosed the financial interest/arrangements with clinical investigators. Statistical analyses were conducted by FDA to determine whether the financial interests/arrangements had any impact on the clinical study outcome. The information provided does not raise any questions about the reliability of the data.

## XII. PANEL MEETING RECOMMENDATION AND FDA'S POST-PANEL ACTION

In accordance with the provisions of section 515(c)(3) of the act as amended by the Safe Medical Devices Act of 1990, this PMA was not referred to the Circulatory Systems Devices Panel, an FDA advisory committee, for review and recommendation because the information in the PMA substantially duplicates information previously reviewed by this panel.

## XIII. CONCLUSIONS DRAWN FROM PRECLINICAL AND CLINICAL STUDIES

The safety and effectiveness of the Resolute Onyx 2.0 mm stent is based on the results of preclinical studies leveraged from the original Resolute Onyx PMA, including biocompatibility, *in vivo* pharmacokinetics (generated on the Resolute product); *in vitro* engineering testing; coating characterization; chemistry, manufacturing and controls information; *in vivo* animal testing; sterilization and stability testing; and a newly

conducted clinical study (RESOLUTE ONYX 2.0 mm). These test results revealed the following:

**A. Effectiveness Conclusions**

The primary endpoint of TLF in the ITT primary analysis set at 12 months was 5.0% (5/100), fulfilling the pre-specified performance criterion (upper 1-sided 95% CI of 10.2%, compared with the performance goal of 19%, $p < 0.001$). Furthermore, the rate of TLF in the ITT worst case analysis set was 5.9% (6/101) (upper 1-sided 95% CI of 11.4%, compared with the performance goal of 19%).

**B. Safety Conclusions**

The risks of the device are based on non-clinical laboratory and animal studies leveraged from the original Resolute Onyx PMA approval, as well as data collected in a clinical study as described above. The biocompatibility, in vivo pharmacokinetics (data generated on the Resolute product), and in vivo performance characteristics of the product provide reasonable assurance of safety and acceptability for clinical use.

The results from the RESOLUTE ONYX 2.0 mm Clinical Study demonstrate that the Resolute Onyx product provides reasonable assurance of safety and effectiveness when used as indicated in accordance with the Instructions for Use (IFU).

**C. Benefit-Risk Determination**

The probable benefits of the device are based on data collected in the clinical study conducted to support PMA Supplement approval as described above. The Resolute Onyx™ 2.0 mm coronary stent has been shown to be beneficial for improving luminal diameter in patients with symptomatic coronary artery disease. The primary end point of Target Lesion Failure (TLF) at 12-months was 5.0% (5/100), fulfilling the pre-specified performance criterion (upper 1-sided 95% CI of 10.2%, compared with the performance goal of 19%, $p < 0.001$). In-segment percent stenosis, as measured by quantitative coronary angiography (QCA), **was 37.92** ± 21.54 (In-Stent 22.49 ± 26.89), In-Segment Binary Stenosis was 20.0% (In-Stent 12.0% ) and In-Segment Late lumen Loss was 0.25 ± 0.41 (In-Stent 0.26 ± 0.48). These results are very good given the small target vessel RVD and not much different than might be seen following current generation DES treatment of disease in larger coronary vessels.

Additional factors to be considered in determining probable risks and benefits for the Resolute Onyx™ 2.0 mm stent system include characterization of the disease, availability of anternative treatments, quality of the study design and conduct, robustness of analysis of study results and risk mitigations. Coronary artery disease (CAD) can be accompanied by symptomatic chest pain or silent ischemia which affects patient's quality of life. CAD is treatable, but if left untreated, the condition can progress to further stenosis within the arteries, increased symptoms and the need

for revascularization. Available treatments for CAD include medical therapy, percutaneous coronary intervention (PCI) and coronary artery bypass graft (CABG) surgery. When treatment for coronary artery disease beyond medications and lifestyle changes is warranted, patients often choose stent deployment over surgical revascularization due to shorter recovery times and the less invasive nature of percutaneous coronary intervention. The risks associated with use of drug eluting stents are already well established, and in comparison to medical therapy, PCI has been shown to reduce the incidence of angina. Patient tolerance of the stent device in the RESOLUTE ONYX 2.0 mm Clinical Study is in line with expectations. The study did not exclude any typical patient subgroups that would be expected to benefit from treatment. Despite the small diameter, the clinical and angiographic results observed in the Resolute Onyx 2.0 mm IDE study are as good as might be expected from a study involving larger diameter drug eluting stents.

1. Patient Perspectives
   This submission did not include specific information on patient perspectives for this device.

In summary, given the available information, the data support the conclusion that for improving coronary luminal diameters in patients, including those with diabetes mellitus, with symptomatic ischemic heart disease due to *de novo* lesions of length ≤ 35 mm in native coronary arteries with reference vessel diameter of 2.0 mm to 5.0 mm, the probable benefits of the Resolute Onyx™ Zotarolimus-Eluting Stent System, including the 2.0 mm stent size, outweigh the probable risks.

### D. Overall Conclusions

The data in this application support the reasonable assurance of safety and effectiveness of the Resolute Onyx™ 2.0 mm Zotarolimus-Eluting Coronary Stent System when used in accordance with the indications for use.

## XIV. CDRH DECISION

CDRH issued an approval order on November 16, 2017. The final conditions of approval cited in the approval order for this panel-track supplement are described below.

1. ODE Lead PMA Post-Approval Study – Continued Follow-up of RESOLUTE ONYX 2.0 mm Clinical Study. The RESOLUTE ONYX (2.0 mm) Clinical Study (G140178) is an open-label, single-arm, multi-center study which enrolled 101 patients and was designed to assess the safety and effectiveness of the Resolute Onyx Zotarolimus Eluting Coronary Stent System (2.0 mm diameter) through 3 years post-index procedure. The primary endpoint is Target Lesion Failure (TLF) at 12 months. Medtronic must collect and report clinical outcomes to FDA through 3 years post-procedure on patients enrolled in the RESOLUTE ONYX 2.0 mm Clinical Study.

2. ODE Lead PMA Post-Approval Study – RESOLUTE ONYX Post-Approval Study (PAS). The RESOLUTE ONYX PAS is a single arm, non-investigational, open-label multi-center study intended to evaluate the safety and effectiveness of the Resolute Onyx Zotarolimus Eluting Coronary Stent Systems in a real world, more-comers population. The study will enroll 510 study subjects: 410 study subjects will be enrolled in the Main Study cohort (2.0 - 4.0 mm) and 100 subjects will be enrolled in an Extra-Large (XLV) sub-study (4.5 - 5.0 mm) cohort. The study will be conducted in up to 25 US sites (representing at least 50% of total enrollment) and up to 5 OUS sites (maximum of 50 study subjects in the Main Study cohort and 49 subjects in the in XLV sub-study cohort). The primary endpoint for all study subjects enrolled in the RESOLUTE ONYX PAS is Target Lesion Failure (TLF) at 12 months, defined as Cardiac Death, Target Vessel Myocardial Infarction or Target Lesion Revascularization. The Main Study cohort will have a performance goal (PG) of 13.2%. The XLV sub-study cohort does not have a formal hypothesis but descriptive statistics will be provided. Medtronic must collect and report clinical outcomes to FDA through 3 years post-procedure on patients enrolled in the RESOLUTE ONYX PAS.

The applicant's manufacturing facilities have been inspected and found to be in compliance with the device Quality System (QS) regulation (21 CFR 820).

## XV. APPROVAL SPECIFICATIONS

Directions for use: See device labeling.

Hazards to Health from Use of the Device: See Indications, Contraindications, Warnings, Precautions, and Adverse Events in the device labeling.

Post-approval Requirements and Restrictions: See approval order.